

GlobalMart Sales Data Analysis — CRISP-DM Report

1. Business Understanding

Business Overview

GlobalMart is a multinational retail and distribution company operating across various regions and offering a wide range of consumer products. The company deals in categories such as Technology, Office Supplies, and Furniture. Due to increased market competition and shifting customer behavior, GlobalMart seeks to leverage data analytics to drive strategic decisions and maintain competitive advantage.

Main Objective

The primary objective of this analysis is to **utilize historical sales data to uncover actionable insights that will help improve profitability, optimize product and customer strategies, and support more informed, data-driven business decisions**. The goal is to translate raw transactional data into meaningful trends, patterns, and recommendations for both operational and strategic planning.

Research Questions

To guide the analysis, the following research questions were formulated:

1. **Which regions and customer segments are the most and least profitable?**
2. **Which product categories and sub-categories contribute the most to sales and profit?**
3. **How do discounts impact profitability across different categories and regions?**
4. **What are the key sales and profit trends over time (monthly, quarterly, yearly)?**
5. **What strategic actions can be taken to improve performance in underperforming areas?**

Success Criteria; will be measured by the ability to uncover actionable insights that inform pricing, inventory, and marketing strategies.

2. Data Understanding

The dataset, sourced from an Excel file provided by the company, includes detailed transactional data with the following fields:

- **Order Date, Ship Date**
- **Country, Region, City**
- **Order ID, Sales, Quantity, Profit**
- **Product Category and Sub-Category**
- **Sales, Quantity, Discount, Profit**

Initial Observations:

- The table from Excel initially had 1010 rows of data.
 - Regions such as *West* and *Central* contribute the most to sales, but show significant variance in profitability.
 - Categories like *Office Supplies* yield the highest sales but with inconsistent profit margins.
 - Discounts heavily impact profit in several regions and categories.
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3. Data Preparation

Transformations and Cleaning:

- Created a robust **Date Table** to support accurate time intelligence.
- Verified consistency of data types and corrected any anomalies (e.g., null values in *Discount & OrderDate*, date mismatches, errors in *OrderDate & Quantity*).
- In some sections with empty values and errors the rows were completely removed from the dataset.
- Built meaningful relationships: Fact Sales (core table) connected to Date, Product and Region dimensions.

Calculated Fields and DAX Measures: (Some of the Measures include)

- `Total Sales = SUM(Sales)`
 - `Total Profit = SUM(Profit)`
 - `Profit Margin = DIVIDE ([Total Profit], [Total Sales])`
 - `YoY Sales Growth = ([This Year Sales] - [Last Year Sales]) / [Last Year Sales]`
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4. Data Modeling and Visualizations

Data Model:

A clean star schema ensures fast queries and easy scalability. However, after close thought, I decided to switch to a snowflake schema with;

- **Sales Table** as the Facts Table.
- **Date, Territory & Products** as the Dimension Tables.
- **Category & Sub-Category Tables** as the Flake Tables.

In total the model has 7 tables including the **Calculations Table** which stores the measures to calculate the various KPIs.

The Tables & their Columns.

Date Table

- Date
- Month Name
- Month Number
- Quarter
- Year
- Quarter
- Week Number
- Day of the Week

Sales Table

- Order Date
- Ship Date
- Product Key
- Quantity
- Sales
- Profit
- Unit Price (Calculated Columns)
- Original Cost (Calculated Columns)
- Days to Ship (Calculated Columns)
- Sub-Category
- Territory Key

Territory Table

- Region
- State
- Territory Key

Products Table

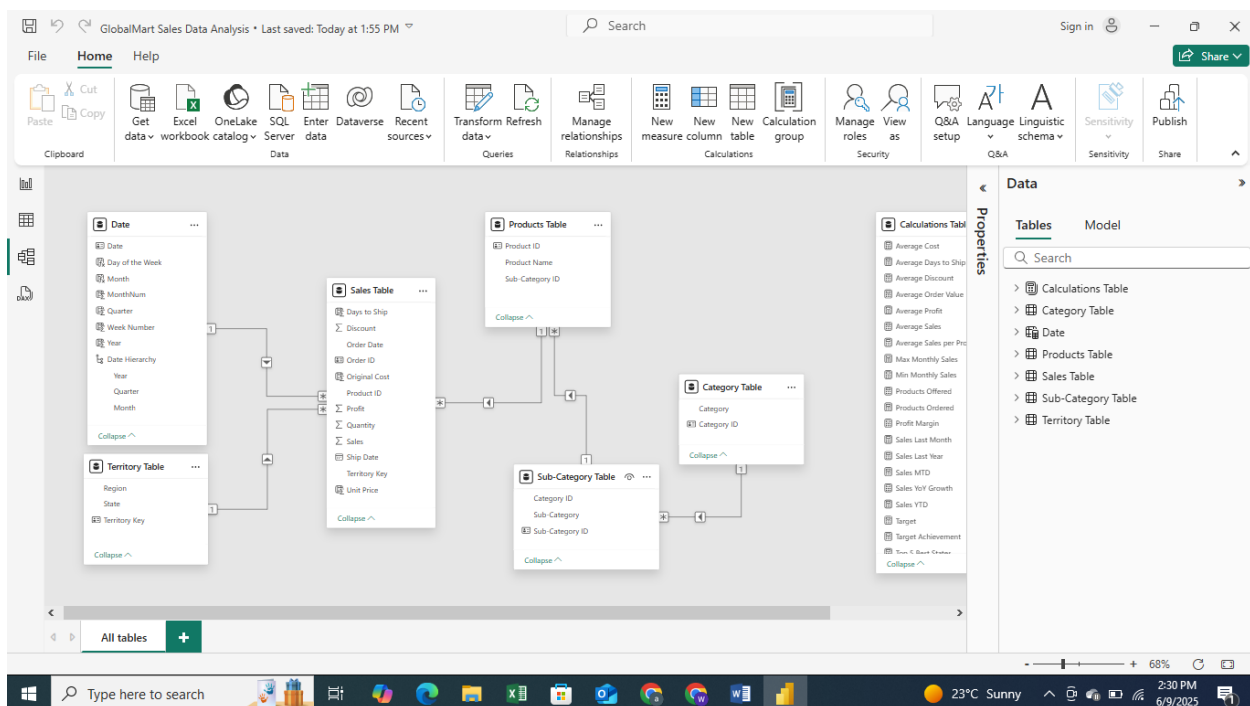
- Product Key
- Product Name

Sub-Category Table

- Sub-Category
- Category Key
- Sub – Category Key

Category Table

- Category
- Category Key



Dashboards Built:

- a) **Sales & Profit Overview**
 - KPIs: Total Sales, Profit, Quantity Sold
 - Time-series visual showing YoY growth and monthly trends
- b) **Regional Analysis**
 - Map visualization: Sales concentration by region
 - *West* had the highest revenue, but *South* showed better profit margins relative to revenue
- c) **Product Analysis**
 - *Office Supplies* drives the most sales but has high discount usage, reducing margins
 - *Technology* showed steady profit with low discount dependency
- d) **Customer Segment Analysis**
 - *Office Supplies & Furniture* have the highest orders.
- e) **Executive Summary**
 - A page showing a brief but detailed summary of major company KPIs across the 3 years from 2022 to 2024.

The report pages have various visualizations from Area Charts, Pie Charts, Column and Bar Graphs among others optimized to provide clear and actionable insights for the company.

The dashboards/reports have buttons, slicers and filters to enable the various department to dig deeper to seek their required information from the visualizations.

5.Actionable Insights & Recommendations (Aligned with Research Questions) - Evaluation

a) Which regions and customer segments are the most and least profitable?

Insight:

- The **West** region generates the highest total sales but also shows the **lowest average profit margin**, likely due to higher discounting or operational costs.
- The **South** region, while mid-level in total revenue, has the **highest profit margin** and consistent customer performance.

Recommendations:

- **Investigate logistics and discounting practices in the West region** to improve profitability.
- Use the South region as a **benchmark for cost efficiency** and apply similar strategies elsewhere.

b) Which product categories and sub-categories contribute the most to sales and profit?

Insight:

- **Office Supplies** products drive the most sales but yield **volatile profits**, likely due to frequent discounts or high procurement costs.
- **Technology**, although contributing less revenue, show **strong and stable profit margins**.
- Within sub-categories, items like **Tables** and **Paper** have high sales with varying profitability.

Recommendations:

- Perform **cost-benefit analysis on Office Supplies discounts** to strike a better balance between sales and margin.
- Expand marketing and stock of **Technology Products** for consistent profit generation.
- **Prioritize procurement planning** for profitable sub-categories with steady demand (e.g., Paper).

c) How do discounts impact profitability across different categories and regions?

Insight:

- A **negative correlation** exists between discount levels and profit margins.

- Discounts over **20%** almost always result in **<10% profit margins** or even losses.

Recommendations:

- Implement **discount ceilings** (e.g., max 15%) for categories with low margin tolerance.
- Use **region-specific discount policies**, focusing on profitability thresholds rather than blanket promotions.
- Create a "**Smart Discount Engine**" that predicts profit impact based on discount inputs.

d) What are the key sales and profit trends over time (monthly, quarterly, yearly)?

Insight:

- Sales consistently **peak in Q4** (Oct–Dec), likely due to end-of-year spending and promotions.
- Profitability, however, **dips slightly in Q4**, suggesting aggressive discounts.
- **Q1 typically sees a slowdown.**

Recommendations:

- **Increase inventory and advertising before Q4**, but manage discounts carefully to protect margin.
- Use **targeted product bundling** instead of high-percentage discounts to retain Q4 profits.
- Launch **early-year promotions (Q1)** to stimulate demand when sales traditionally lag.

e) What strategic actions can be taken to improve performance in underperforming areas?

Insight:

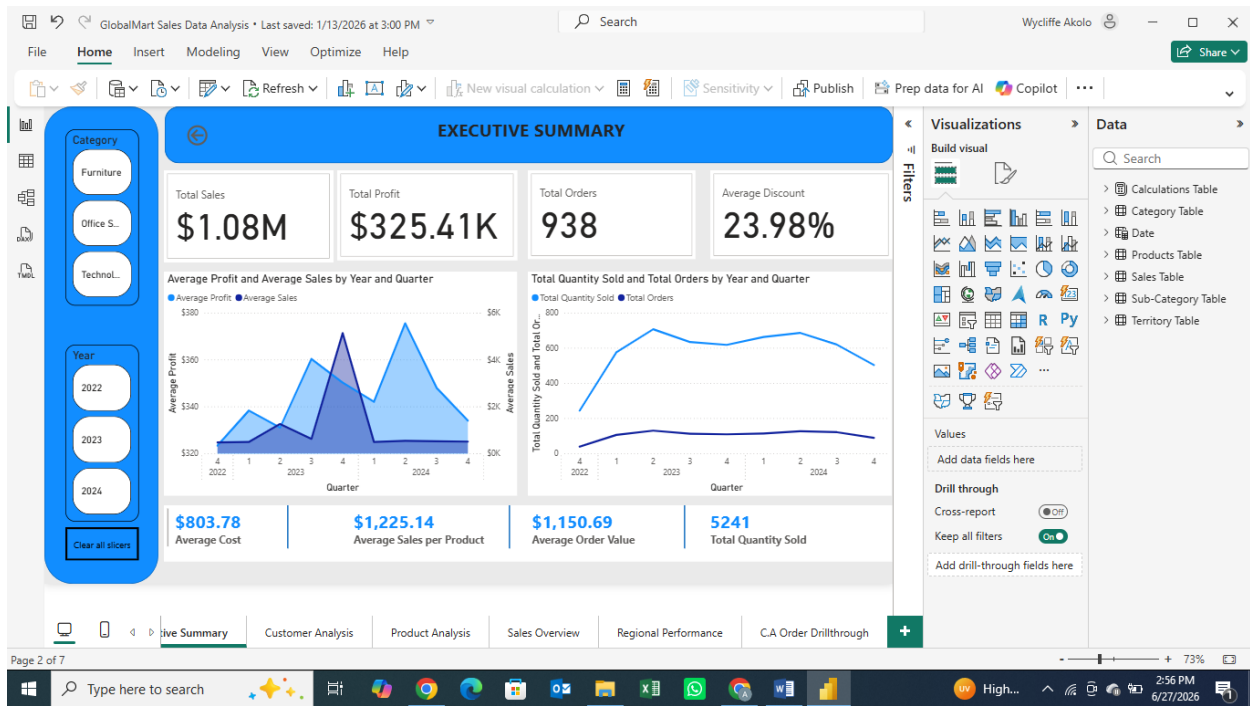
- Underperforming regions tend to have **high shipping lead times** and **greater discount dependency**.
- Some sub-categories have poor sales and profit performance across multiple regions.

Recommendations:

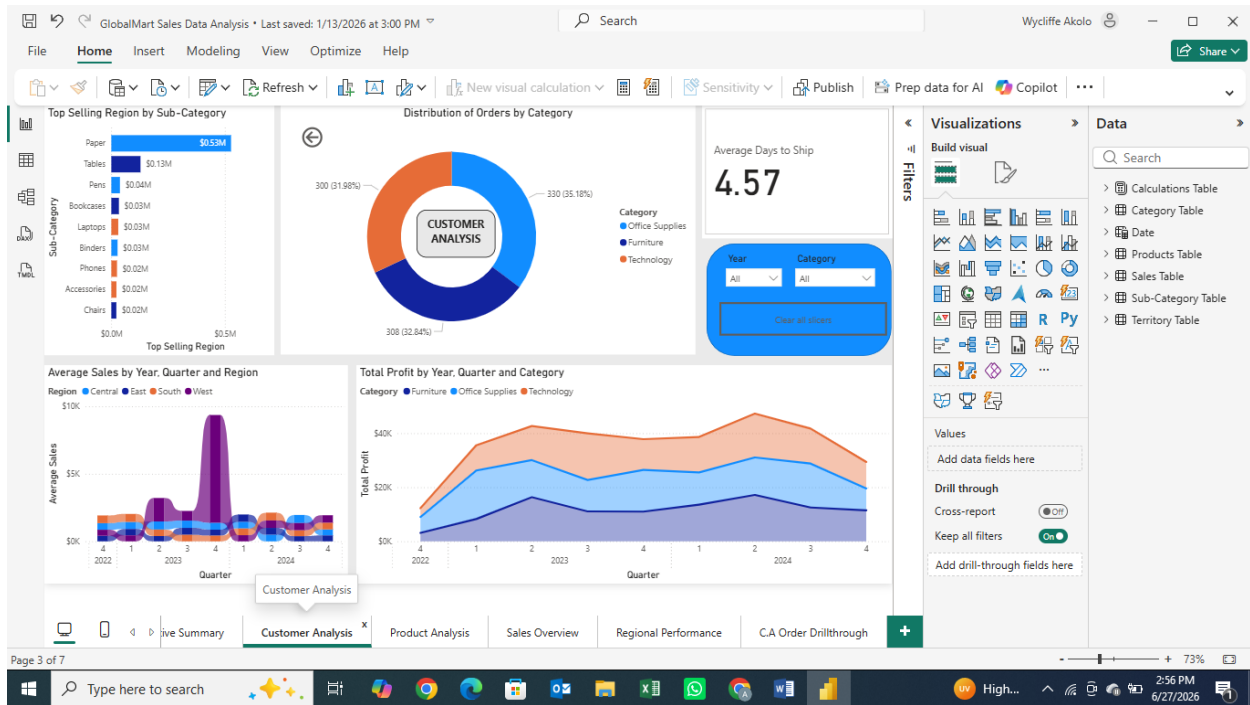
- **Audit regional shipping operations** to reduce delays and cost.
 - Review **product lifecycle** for low-performing items; consider discontinuing or replacing them.
 - Offer **localized product recommendations** to tailor strategies per region.
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SCREEN SHOTS OF THE ANALYSIS IN POWER BI

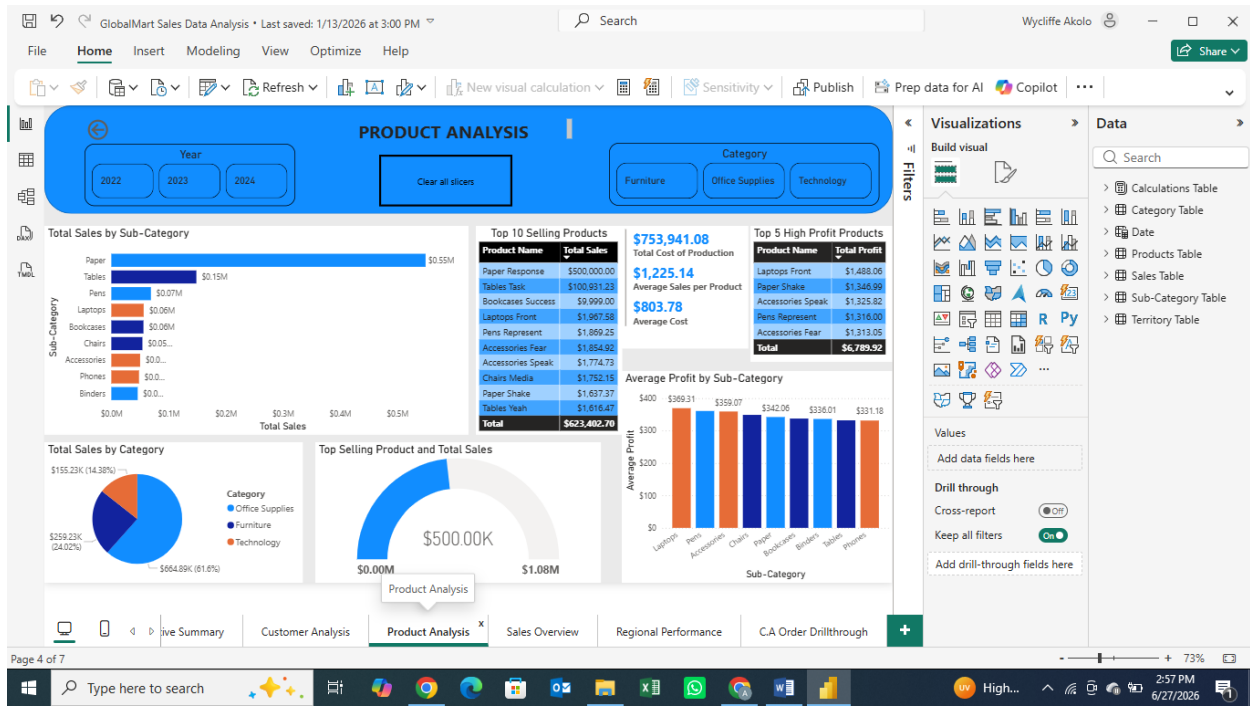
1. EXECUTIVE SUMMARY



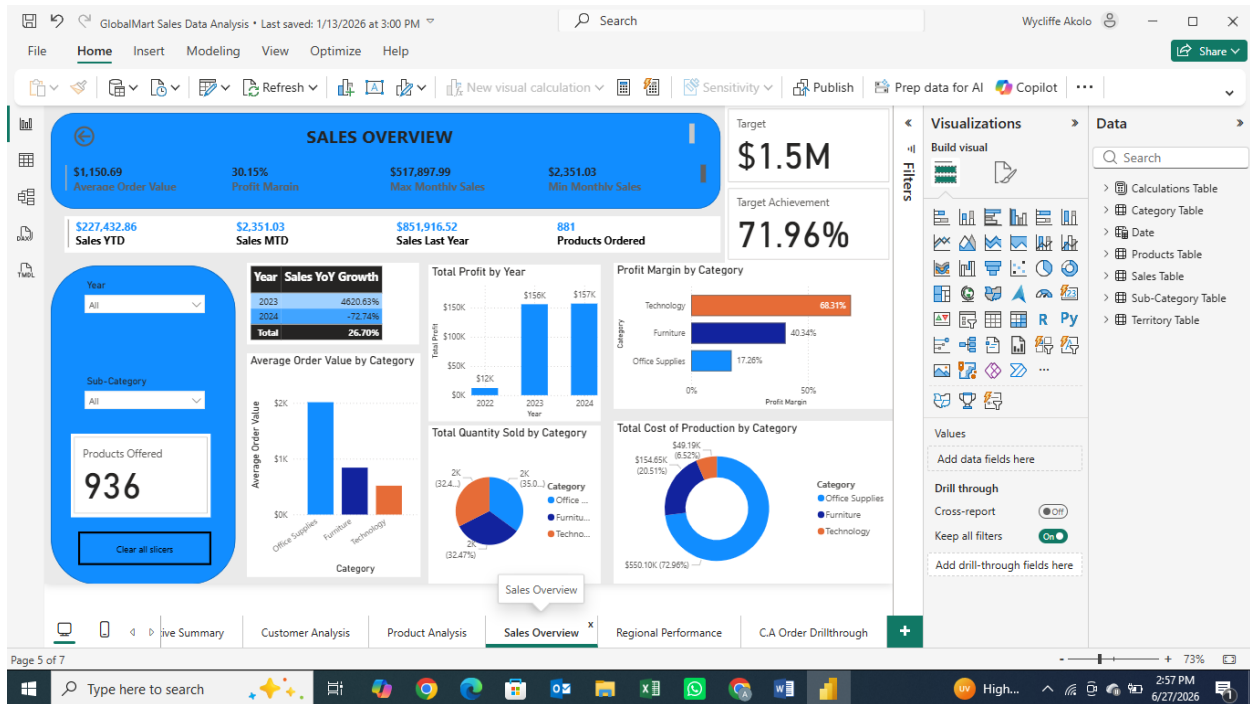
2. CUSTOMER ANALYSIS



3. PRODUCT ANALYSIS



4. SALES OVERVIEW



5. REGIONAL PERFORMANCE

